CP115
Computer Programming 1
Semester I
Session 2026/2027
2 hours

ALL OUT ARENA

CP115
Pengaturcaraan Komputer 1
Semester I
Sesi 2026/2027
2 jam

No. Matrik *Matric No.*	**No. Kad Pengenalan** *Identity Card No.*	**Nama** *Name.*
		Class:

(*Isikan maklumat dengan lengkap*)

KOLEJ MATRIKULASI KEDAH
KEDAH MATRICULATION COLLEGE

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
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Prepared by:

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(HASNI BINTI ABDUL GHANI)

Untuk Kegunaan Pemeriksa *For Examiner's Use*		
No. Soalan Question No.	Markah *Mark*	
	Pemeriksa *Examiner*	KP / KKP
1 (25)		
2 (12)		
3 (13)		
4 (8)		
5 (14)		
6 (18)		
JUMLAH *TOTAL*		

Kertas soalan ini mengandungi **6** halaman bercetak.
This question paper consists of **6** *printed pages.*
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QUESTION 1 [25 MARKS]

(a) Explain first THREE steps in the problem-solving process that a programmer would perform when developing **a student attendance system**. **[6 marks]**

(b) Identify whether each of the following real-world system operations represents an **Input, Process,** or **Output**. **[4 marks]**

System Operation	Operation
Reading an atmospheric temperature value from an IoT digital sensor chip.	
Evaluating if a user's password meets mandatory security complexity rules.	
Generating an automated PDF tax invoice receipt for a completed transaction.	
Converting a recorded voice audio stream into formatted text tokens.	

(c) A parking kiosk calculates total charges based on entry and exit hours entered by the user. The rate is RM 3.00 for the first hour and RM 2.00 for each subsequent hour. An 8% service tax is added to the total fee before displaying the final payable amount. Write the IPO Model. **[4 marks]**

Input	
Process	
Output	

(d) A college wants to develop a system to calculate the amount a student must pay for printing. The user enters the number of black-and-white pages and colour pages. Black-and-white printing costs RM0.10 per page and colour printing costs RM0.60 per page. Write an IPO Model to calculates and displays the total charge. **[3 marks]**

Input	
Process	
Output	

(e) An automated billing system calculates a customer's monthly payment based on total kilowatt-hours (kWh) consumed. If total consumption is 200 kWh or less, the rate is RM 0.20 per kWh. If total consumption exceeds 200 kWh, the first 200 kWh are charged at RM 0.20 per kWh, and all additional units above 200 kWh are charged at RM 0.35 per kWh. Write an IPO Model to displays the final total bill amount. **[4 marks]**

Input	
Process	
Output	

(f) A fitness app continuously prompts the user to enter daily step counts until a sentinel value of -1 is entered. The system calculates and displays the total accumulated steps and the overall average daily step count across all valid entries. **[4 marks]**

Input	
Process	
Output	

QUESTION 2 [12 MARKS]

(a) Compare an **Algorithm, Pseudocode,** and **Flowchart**. **[3 marks]**

_____

_____

_____

(b) Study the pseudocode below and convert it into a flowchart. Use the correct flowchart symbols.

[6 marks]

```
START
  Set total = 0
  Set count = 0
  Read quantity
  WHILE quantity != -99
    IF quantity > 0 THEN
      Set total = total + quantity
      Set count = count + 1
    ENDIF
    Read quantity
  ENDWHILE
  IF count > 0 THEN
    Set average = total / count
    Print average
  ELSE
    Print "No items processed"
  ENDIF
 STOP
```

(c) Write pseudocode for a program that accepts the length and width of a rectangular garden. The program calculates and displays the area and perimeter of the garden. **[3 marks]**

QUESTION 3 [13 MARKS]

(a) Programming languages utilize system translators to convert instructions into machine code.

(i) Differentiate between a **Compiler** and an **Interpreter**. **[2 marks]**

_____

_____

_____

(ii) State **TWO (2)** main programming paradigms used in modern software development. **[1 mark]**

(b) For each identifier below, state whether it is VALID or INVALID. Give a reason for each invalid **[2 marks]**

(i) `student_name` -

(ii) `2ndMark` -

(iii) `total-cost` -

(iv) `class` -

(c) Identify the data type of each value: **[2 marks]**

(i) 18 -

(ii)6.25 -

(iii)CP115 -

(iv) False -

(d) Write one suitable Python statement for each requirement. **[3 marks]**

(i) Read a whole number from the user and store it in a variable named `number_of_items`.

(ii) Calculate the square of variable length and assign the result to area.

(iii) Display the message "Calculation completed" followed by the value of area.

(e) Determine the output of the following Python code. Show the value of each variable printed. **[3 marks]**

`a = 17`
`b = 5`
`x = a // b + 2 ** 3`
`y = a % b * 2`
`z = (a > 10) and (b == 4)`
`print(x)`
`print(y)`
`print(z)`	

QUESTION 4 [8 MARKS]

(a) Explain what is meant by a sequence control structure. **[1 mark]**

(b) A sports centre charges RM12.00 for each adult ticket and RM7.50 for each student ticket. A customer buys adult_count adult tickets and student_count student tickets. Write a Python program that displays both category totals and the total payment in a clear format **[4 marks]**

(c) Categorize each programming error scenario as either **Syntax Error, Run-time Error,** or **Logic Error**. **[3 marks]**

(i) Missing closing parenthesis ')' at the end of a print statement. _____

(ii) Computing average test marks using formula: avg = a + b + c / 3. _____

(iii) Attempting to divide a total score variable by zero during execution. _____

QUESTION 5 [14 MARKS]

(a) State the selection control structure that is most suitable for each situation: single selection, dual selection or multiple selection. **[3 marks]**

(i) Display “Eligible” only when a student's attendance is at least 80%. _____

(ii) Display “Pass” if a mark is at least 50%; otherwise display “Fail”. _____

(iii) Display a different grade message for each of the grades A, B, C, D and F. _____

(b) Write the Python Boolean expression for each condition. **[2 marks]**

(i) score is from 40 to 69 inclusive.

(ii) age is at least 18 and citizenship is equal to "Malaysian".

(c) Write a Python program that accepts a student's mark. The program displays “Invalid mark” if the mark is below 0 or above 100. Otherwise, it displays “Pass” if the mark is at least 50, or “Fail” if the mark is below 50. **[6 marks]**

(d) Complete the Python program below to calculate worker overtime pay. Hours above 40 are paid at 1.5 times the hourly rate. **[3 marks]**

```
hours = float(input("Enter total hours worked: "))
rate = float(input("Enter hourly rate: "))

if _____________________:  # Fill in condition
    overtime_hours = hours - 40
    pay = ___________________  # Fill in calculation
else:
    pay = ___________________  # Fill in calculation

print("Total Pay: RM", pay)
```

QUESTION 6 [18 MARKS]

(a) Rewrite the following while loop structure into an equivalent Python program using a `for loop with range()`. **[4 marks]**

```
count = 25
total = 0

while count >= 5:
    total = total + count
    print("Value:", count)
    count = count - 5

print("Total Sum:", total)
```

(b) Write TWO separate Python code segments to perform the following tasks. Use the loop type specified in each task.

(i) Using a for loop with range(), display the multiples of 4 from 4 to 40 inclusive. **[3 marks]**

(ii) Using a while loop, display 25, 20, 15, 10 and 5, in that order. **[3 marks]**

(c) Write a complete Python program that repeatedly prompts the user to enter student examination marks until a sentinel value of **-1** is entered. **[8 marks]**

The program must perform the following:

1. Accept student marks between 0 and 100. Display error for invalid marks.
2. Calculate and display total number of valid students tested.
3. Calculate and display overall average mark of the class.
4. **Calculate and display total number of students who need consultation (marks less than 50).**
5. Determine and display the name of the student who achieved the highest mark, along with their score.

END OF QUESTION PAPER